Topic 4 — Linear Regression, Correlation, and Logistic Regression

Linear Regression: Motivation

A **linear regression** fits a straight line to observed data to demonstrate a linear relationship between the variables, and it makes predictions on new data.

Example

Exam marks vs. revision hours: this can be represented by a scatter plot, with a fitted line drawn through the data.

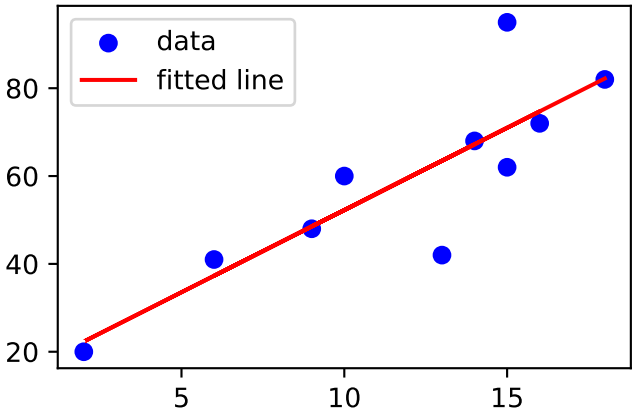


Figure 1

The Method of Least Squares

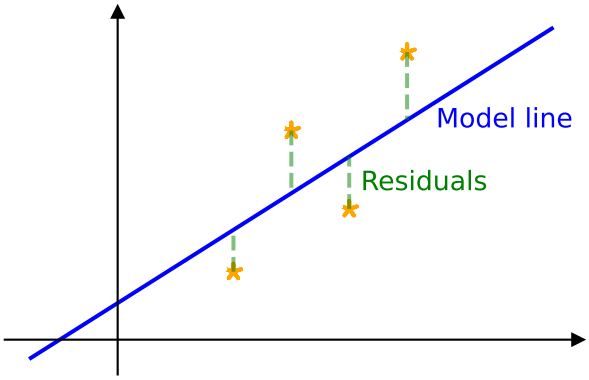


Figure 2

Assuming data pairs $(x_1, y_1), \ldots, (x_n, y_n)$, the model line takes the form $\mathbb{E}(y) = \beta_0 + \beta_1 x$, where y is the **dependent (response) variable** and x is the **independent (explanatory/predictor) variable**. The variables are related by

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where ϵ is the error. We look for the line minimising the sum of squared residuals

$$f(\beta_0, \beta_1) = \sum_{i=1}^{n} (\beta_0 + \beta_1 x_i - y_i)^2.$$

Least Squares Estimates

Definition

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}};$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Note that

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2.$$

The least squares model is therefore

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Correlation

Definition

The strength of a linear relationship between the variables can be measured by the **Pearson correlation coefficient**:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

Remark

r can take values between -1 and 1, where -1 and 1 represent a perfect linear relationship, and $r = 0$ means no linear relationship. A positive r denotes a positive correlation, a negative r a negative correlation. As a rule of thumb:

$$\begin{aligned} |r| &> 0.7 && \text{Strong correlation} \\ 0.7 \geq |r| &> 0.4 && \text{Moderate correlation} \\ |r| &\leq 0.4 && \text{Weak correlation} \end{aligned}$$

Example: Revision Hours vs. Exam Marks

Example

$\bar{x} = 11.8$, $\bar{y} = 59$.

Revision Hours (x_i)	18	2	13	14	6	15	16	9	10	15	**Sum**
Mark (y_i)	82	20	42	68	41	95	72	48	60	62	
$x_i - \bar{x}$	6.2	-9.8	1.2	2.2	-5.8	3.2	4.2	-2.8	-1.8	3.2	0
$y_i - \bar{y}$	23	-39	-17	9	-18	36	13	-11	1	3	0
$(x_i - \bar{x})(y_i - \bar{y})$	142.6	382.2	-20.4	19.8	104.4	115.2	54.6	30.8	-1.8	9.6	837
$(x_i - \bar{x})^2$	38.44	96.04	1.44	4.84	33.64	10.24	17.64	7.84	3.24	10.24	223.6
$(y_i - \bar{y})^2$	529	1521	289	81	324	1296	169	121	1	9	4340

Example

$$\hat{\beta}_1 = \frac{837}{223.6} = 3.743, \qquad \hat{\beta}_0 = 59 - 3.743 \times 11.8 = 14.829.$$

Regression line: $\hat{y} = 3.743x + 14.829$.

$$r = \frac{837}{\sqrt{223.6}\sqrt{4340}} = 0.85,$$

indicating a strong positive correlation. $r^2 = 0.723$: 72.3% of the variance of the marks is explained by revision. The regression model fits the data well.

Example

For 11 hours of revision:

$$\hat{y} = 3.743 \times 11 + 14.829 \approx 56.$$

Common sense should be used when predicting: we cannot predict outside the possible range of x values (e.g. -5 hours of revision makes no sense).

Multiple Linear Regression

This extends the simple linear regression model $y = \beta_0 + \beta_1 x + \epsilon$ to several independent variables $x_1, \dots, x_n$ and a single dependent variable y:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon.$$

We concentrate on $n = 2$ independent variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon.$$

Example. x_1 — hours of revision, x_2 — number of attended lectures, y — mark.

Coding the variables:

$$v_i = y_i - \bar{y}, \qquad u_{1i} = x_{1i} - \bar{x}_1, \qquad u_{2i} = x_{2i} - \bar{x}_2,$$

we write the model as $v = \beta_0' + \beta_1 u_1 + \beta_2 u_2 + e$, and minimise

$$Q = \sum_{i=1}^{n} (v_i - \beta_0' - \beta_1 u_{1i} - \beta_2 u_{2i})^2.$$

This leads to the **normal equations**. Using

$$S_{pq} = \sum_{i=1}^{n} (x_{pi} - \bar{x}_p)(x_{qi} - \bar{x}_q), \qquad S_{0p} = \sum_{i=1}^{n} (x_{pi} - \bar{x}_p)(y_i - \bar{y}), \qquad p, q = 1, 2,$$

they become

$$\hat{\beta}_1 S_{11} + \hat{\beta}_2 S_{12} = S_{01}, \qquad \hat{\beta}_1 S_{12} + \hat{\beta}_2 S_{22} = S_{02}.$$

With $D = S_{11} S_{22} - S_{12}^2$,

$$\hat{\beta}_1 = \frac{S_{22} S_{01} - S_{12} S_{02}}{D}, \qquad \hat{\beta}_2 = \frac{S_{11} S_{02} - S_{12} S_{01}}{D}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2,$$

giving the model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$.

Logistic Regression: Motivation

The main use of logistic regression is to predict a binary outcome from a linear combination of independent variables — e.g. predicting whether a student passes an exam from hours of revision. Here the data is binary, taking only values 0 and 1 (formally, the dependent variable $Y \sim Bin(1, p)$), and we want to predict p.

Example: Pass/Fail vs. Revision Hours

Example

A student spent various numbers of hours revising, and we record whether they passed (1) or failed (0):

Hours	10	11	12	13	14	17	18	19	20
P/F	0	0	0	1	0	1	1	0	1

A scatter plot of this data (points at $y = 0$ or $y = 1$ for each hour value) is naturally fitted with an S-shaped (sigmoid) curve rising from near 0 for low hours to near 1 for high hours.

The Logistic (Sigmoid) Model

Definition

The logistic regression model chooses a fitting curve of the form

$$f(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}, \qquad (1 + e^{-(\beta_0 + \beta_1 x)}) > 0,$$

where β_0, β_1 are fitting parameters estimated from the data. Writing $y = \beta_0 + \beta_1 x$ and $g(y) = \dfrac{1}{1+e^{-y}} = \dfrac{e^y}{1+e^y}$, we have $f(x) = g(\beta_0 + \beta_1 x)$.

Always $0 < f(x) < 1$. If $\beta_1 > 0$, then $\lim\limits_{x \to +\infty} f(x) = 1$ and $\lim\limits_{x \to -\infty} f(x) = 0$. Indeed,

$$\frac{1}{1 + e^{-\beta_1 x - \beta_0}} = \frac{1}{1 + \dfrac{e^{-\beta_0}}{e^{\beta_1 x}}} \to \frac{1}{1+0} = 1 \qquad (x \to +\infty,\ \beta_1 > 0),$$

$$\frac{1}{1 + e^{-\beta_1 x} e^{-\beta_0}} \to 0 \qquad (x \to -\infty).$$

Also, $f(x)$ is increasing.

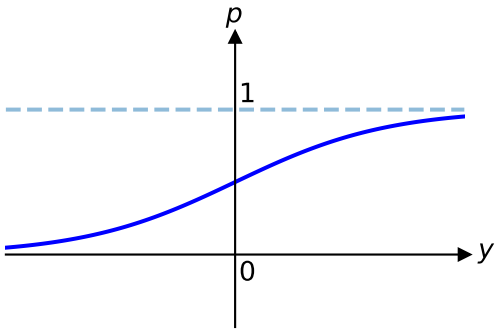


Figure 3

Deriving the Logit Function

We use this curve for prediction: $g(y) = p \in [0, 1]$ is the probability (e.g. of passing). Solving for y:

$$g(y) = \frac{e^y}{1+e^y} = p \implies e^y = p + pe^y \implies e^y(1-p) = p \implies e^y = \frac{p}{1-p} \implies y = \ln \frac{p}{1-p}.$$

Definition

The quantity $\dfrac{p}{1-p}$ is called the **odds**.

Definition

The natural logarithm of the odds is called the **logit function**:

$$\text{logit}(p) = \ln \frac{p}{1-p}, \qquad 0 < p < 1.$$

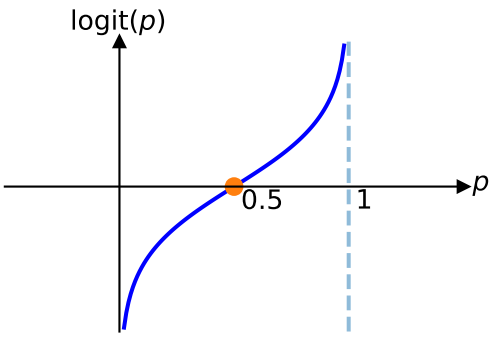


Figure 4

So $g(y) = p$ means $y = \text{logit}(p)$: the logit is the inverse function of the logistic function $g(y) = \dfrac{1}{1+e^{-y}} = \dfrac{e^y}{1+e^y}$.

Several Predictor Variables

If the data depends on several predictor variables $x_1, x_2, \dots, x_n$ (e.g. x_1 = hours of revision, x_2 = cups of coffee, …), we still consider

$$f(x_1, \dots, x_n) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}},$$

with $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$, so that $f(x_1, \dots, x_n) = g(y) = p \in (0, 1)$, and

$$\hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_n x_n = y = \text{logit}(p),$$

where $\hat{}$ denotes estimated parameters.